

Question	Answer	Marks
7(a)	• (electron) neutrino • (electron) antineutrino • electron • positron Any two, 1 mark each	B2
7(b)	only meson <u>and</u> neutron are underlined	B1
7(c)	(quark composition is) up, up, down / uud (charge =) $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = (+1)e = (+) 1.6 \times 10^{-19} \text{ (C)}$	B1
	or (charge =) $1.07 \times 10^{-19} + 1.07 \times 10^{-19} - 5.33 \times 10^{-20} = (+) 1.6 \times 10^{-19} \text{ (C)}$	A1